

TOPOLOGICAL GAMES AND BAIRE SPACES

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For the UT San Antonio Mathematics student research seminar

These notes study *topological games*: infinite games in which two players alternately choose open sets, and in which topological properties of a space are characterized by which player has a winning strategy. The two main results are that a space is a Baire space exactly when the first player has no winning strategy in the Banach–Mazur game (see Theorem 2.3 below), and that a metrizable space is completely metrizable exactly when the second player has a winning strategy in the strong Choquet game (see Theorem 3.4 below). The easy half of the second result, together with the first, already gives a game-theoretic proof of the Baire category theorem (see Corollary 3.5 below). The hard half, that a winning strategy for the second player produces a complete metric, is proved in Section 4; it uses three classical tools that are worth knowing for their own sake: Alexandrov’s theorem on G_δ sets, Stone’s theorem on locally finite refinements, and König’s lemma.

0. BAIRE CATEGORY

In this section we recall the notions of category and show that a space is a Baire space exactly when no non-empty open subset of the space is of the first category (see Theorem 0.3 below).

Definition 0.1. Let X be a topological space, and let $E \subseteq X$ be given. The closure of E will be denoted by $\overline{E}$, and the interior of E by E° .

- (1) The set E is said to be *nowhere dense* in X if $(\overline{E})^\circ = \emptyset$. Equivalently, the set E is nowhere dense if every non-empty open set $U \subseteq X$ contains a non-empty open set V such that $V \cap E = \emptyset$.
- (2) The set E is said to be of the *first category* in X (or *meager*) if E is the union of a countable family of nowhere dense sets. A set which is not of the first category is said to be of the *second category* (or *non-meager*).
- (3) A set $C \subseteq X$ is said to be *comeager* (or *residual*) if the complement $X \setminus C$ is of the first category.
- (4) The space X is called a *Baire space* if the intersection of every countable family of dense open subsets of X is dense in X .



Remark 0.2. The notions in Definition 0.1 classify subsets according to topological “size,” providing a qualitative analogue of measure theory.

- (1) A nowhere dense set is thin everywhere: every non-empty open set contains a non-empty open set that misses the nowhere dense set entirely. Every finite subset of $\mathbb{R}$ is nowhere dense, and so is the Cantor ternary set.
- (2) A set of the first category is topologically negligible, and a countable union of such sets is again of the first category, just as a countable union of Lebesgue-null sets is null. The rationals $\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\}$ form a set of the first category in $\mathbb{R}$. The analogy has limits: the line $\mathbb{R}$ is the union of a set of the first category and a set of Lebesgue measure zero [Oxt80, Chapter 1], so “small” in one sense can be “large” in the other.
- (3) A comeager set contains “most” points of the space; the elements of a comeager set are called the *generic* points of X . In $\mathbb{R}$, the set of irrational numbers is comeager.

- (4) In Theorem 0.3 below we show that X is a Baire space exactly when no non-empty open subset of X is negligible. This *local* condition is stronger than asking that X itself be of the second category in X : the disjoint union $X = (\mathbb{Q} \times \{0\}) \cup (\mathbb{R} \times \{1\}) \subseteq \mathbb{R}^2$ is of the second category in itself, because the open subset $\mathbb{R} \times \{1\}$ of X is, but X is not a Baire space, because the open subset $\mathbb{Q} \times \{0\}$ of X is of the first category (see Figure 1).



FIGURE 1. The space $X = (\mathbb{Q} \times \{0\}) \cup (\mathbb{R} \times \{1\})$ of Remark 0.2(4). Each line is an open subset of X . The upper line makes X of the second category in itself; the lower line, a countable union of points, is a non-empty open set of the first category, so X is not a Baire space.

Theorem 0.3. *A topological space X is a Baire space if and only if no non-empty open subset of X is of the first category.*

Proof. Suppose that X is a Baire space, and let $G \subseteq X$ be open and non-empty. If $G = \bigcup_{n=1}^{\infty} E_n$ with each E_n nowhere dense, then $V_n = X \setminus \overline{E_n}$ is open and dense in X . Hence $\bigcap_{n=1}^{\infty} V_n$ is dense in X , so $\bigcap_{n=1}^{\infty} V_n$ must intersect G . However, we have

$$G \cap \bigcap_{n=1}^{\infty} V_n = G \setminus \bigcup_{n=1}^{\infty} \overline{E_n} = \emptyset,$$

a contradiction. Thus G is of the second category.

Conversely, suppose no non-empty open subset of X is of the first category. Let $\{V_n\}$ be a sequence of dense open subsets of X , and let $W \subseteq X$ be open and non-empty. Write $E_n = X \setminus V_n$. Each E_n is closed, so $\overline{E_n} = E_n$, and E_n has empty interior because V_n is dense; thus each E_n is nowhere dense. If $W \cap \bigcap_{n=1}^{\infty} V_n = \emptyset$, then $W \subseteq \bigcup_{n=1}^{\infty} E_n$. This makes W a non-empty open set of the first category, contrary to our hypothesis. Hence $W \cap \bigcap_{n=1}^{\infty} V_n \neq \emptyset$, and the intersection is dense. ☺

1. INFINITE GAMES

In this section we fix the formal framework for the games of these notes: trees of positions, plays, payoff sets, and strategies.

Definition 1.1. Let A be a non-empty set. We denote by $A^{<\omega}$ the collection of all finite sequences $s = (a_0, \dots, a_{n-1})$ of elements of A . The length n of s is denoted by $|s|$; the empty sequence $\emptyset$ has length 0. If $t \in A^{<\omega}$ and $|t| \leq |s|$ with $t_i = s_i$ for $0 \leq i < |t|$, we write $t \subseteq s$. For $s \in A^{<\omega}$ and $a \in A$, we write $s \frown a = (s_0, \dots, s_{|s|-1}, a)$ for the extension of s by a .

A non-empty subset $T \subseteq A^{<\omega}$ is called a *tree* if $s \in T$ and $t \subseteq s$ implies $t \in T$. For each $s \in T$, we write

$$\text{Succ}(s) = \{a \in A : s \frown a \in T\}.$$

We assume throughout that $\text{Succ}(s) \neq \emptyset$ for every $s \in T$. An infinite sequence $x = (a_n)_{n=0}^{\infty} \in A^{\omega}$ is a *play* in T if $(a_0, \dots, a_{n-1}) \in T$ for all $n \geq 1$. The set of all such plays is denoted by $[T]$.

Definition 1.2. An *infinite game* is a pair (T, W) , where $T \subseteq A^{<\omega}$ is a tree and $W \subseteq [T]$. Two players, I and II, alternately choose elements $a_n \in A$ such that $(a_0, \dots, a_n) \in T$. Player I chooses a_{2n} ; Player II chooses a_{2n+1} ($n = 0, 1, 2, \dots$). Player II *wins* the resulting play $x = (a_n) \in [T]$ if $x \in W$; otherwise, Player I wins. The elements of T are called *positions*.

A *strategy* for Player I is a function σ defined on the positions of even length such that $\sigma(s) \in \text{Succ}(s)$. A play or position is *consistent* with σ if $a_{2n} = \sigma(a_0, \dots, a_{2n-1})$ whenever a_{2n} is defined. We say that σ is a *winning strategy* for Player I if every play consistent with σ lies in $[T] \setminus W$.

A strategy τ for Player II and the notion of a winning strategy for Player II are defined analogously.



Remarks 1.3. (1) In these notes W is the set of plays won by Player II. Many texts, including [Kec95], use the opposite convention, in which the payoff set is Player I's. Ours suits the games below, where Player II is the player who wants a non-empty intersection.
 (2) The two players need not choose from the same kind of object: if Player I chooses from a set A_I and Player II from a set A_{II} , take $A = A_I \cup A_{II}$ and let the tree T record which moves are legal.
 (3) At most one player has a winning strategy: if σ and τ were winning strategies for Players I and II, the unique play consistent with both would lie both outside and inside W .

2. THE BANACH–MAZUR GAME

In this section we define the Banach–Mazur game and show that a space is a Baire space exactly when the first player has no winning strategy (see Theorem 2.3 below).

Definition 2.1. Let X be a topological space, and let $\mathcal{O}^+(X)$ be the collection of all non-empty open subsets of X . The *Banach–Mazur game* on X , denoted by $\text{BM}(X)$, is the game (T, W) where:

- (1) The set of moves is $A = \mathcal{O}^+(X)$.
- (2) The tree T consists of all finite sequences $U_0 \supseteq V_0 \supseteq U_1 \supseteq V_1 \supseteq \dots$ of sets in $\mathcal{O}^+(X)$.
- (3) The payoff set is $W = \{(U_0, V_0, U_1, \dots) \in [T] : \bigcap_{n=0}^{\infty} U_n \neq \emptyset\}$.

Thus Player I plays the sets U_n , Player II plays the sets V_n , and Player II wins if the sets played have a point in common.

Example 2.2. (1) In $\text{BM}(\mathbb{Q})$, Player I has a winning strategy. Fix an enumeration $q_0, q_1, \dots$ of $\mathbb{Q}$, and let Player I play $U_0 = \mathbb{Q} \setminus \{q_0\}$ and $U_n = V_{n-1} \setminus \{q_n\}$ for $n \geq 1$. These sets are open and non-empty, and every q_n is missing from U_n , so $\bigcap_n U_n = \emptyset$ (see Figure 2).
 (2) In $\text{BM}(\mathbb{R})$, Player II has a winning strategy: in answer to U_n , she plays a bounded open interval V_n with $\overline{V_n} \subseteq U_n$. The compact sets $\overline{V_n}$ are nested, so these sets have a common point, and that point lies in every U_n .

Theorem 2.3. Let X be a topological space.

- (1) Player I has a winning strategy in $\text{BM}(X)$ if and only if X contains a non-empty open set of the first category.
- (2) The space X is a Baire space if and only if Player I has no winning strategy in $\text{BM}(X)$.

Proof. By Theorem 0.3, part (2) is an immediate consequence of part (1).

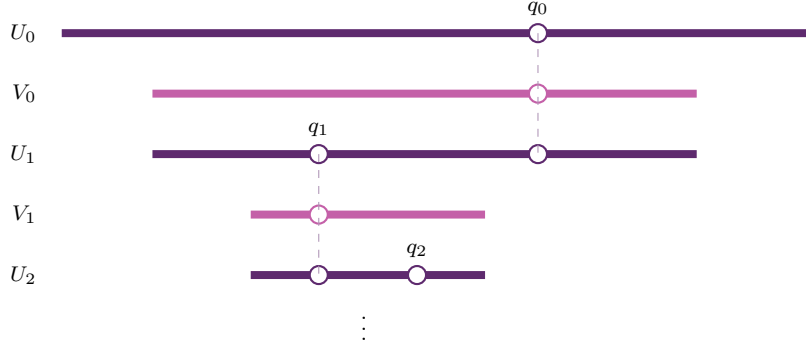


FIGURE 2. Player I wins $\text{BM}(\mathbb{Q})$ (see Example 2.2(1)). The moves of Player I are drawn in plum and those of Player II in pink, as intervals of rationals. At step n Player I removes the rational q_n (a hole), and a hole, once made, persists in every later set. Every rational is removed at some step, so the sets have no point in common.

Suppose first that $G \subseteq X$ is open, non-empty, and of the first category. Then $G = \bigcup_{n=0}^{\infty} E_n$, where each E_n is nowhere dense. Player I plays $U_0 = G$. If Player II responds with $V_0 \subseteq U_0$, the set $V_0 \setminus \overline{E_0}$ is open and non-empty. Player I may therefore choose a non-empty open set $U_1 \subseteq V_0 \setminus \overline{E_0}$. In general, given V_{n-1} , Player I chooses an open set

$$U_n \subseteq V_{n-1} \setminus \overline{E_{n-1}}, \quad U_n \neq \emptyset.$$

This defines a strategy σ for Player I. For any play consistent with σ , we have $U_n \cap E_{n-1} = \emptyset$ for all $n \geq 1$, so that

$$\bigcap_{n=0}^{\infty} U_n \subseteq G \setminus \bigcup_{n=0}^{\infty} E_n = \emptyset.$$

Thus σ is a winning strategy for Player I.

Conversely, suppose Player I has a winning strategy σ , and put $U_0 = \sigma(\emptyset)$. We show that U_0 is of the first category. For a position p of odd length $2n+1$ consistent with σ , we shall write U^p for the last entry of p , the set U_n played by Player I.

We build, for each $n \geq 0$, a family $\mathcal{P}_n$ of positions of length $2n+1$ consistent with σ , such that:

- (a) the sets U^p , $p \in \mathcal{P}_n$, are pairwise disjoint;
- (b) their union $D_n = \bigcup_{p \in \mathcal{P}_n} U^p$ is dense in U_0 ;
- (c) every $p \in \mathcal{P}_{n+1}$ extends some $q \in \mathcal{P}_n$.

Let $\mathcal{P}_0 = \{(U_0)\}$. Given $\mathcal{P}_n$, fix $q \in \mathcal{P}_n$. Each non-empty open $V \subseteq U^q$ is a legal move for Player II at q , and leads to the position $q \frown V \frown \sigma(q \frown V)$ of length $2n+3$. By Zorn's lemma, choose a family $\mathcal{P}(q)$ of such positions, maximal with respect to the property that their last entries are pairwise disjoint. The union of these last entries is dense in U^q : otherwise some non-empty open $W \subseteq U^q$ misses all of them, and then the position $q \frown W \frown \sigma(q \frown W)$, whose last entry lies in W , could be added to $\mathcal{P}(q)$, contradicting maximality. Put $\mathcal{P}_{n+1} = \bigcup_{q \in \mathcal{P}_n} \mathcal{P}(q)$. Property (c) holds by construction. Property (a) holds because last entries of positions in the same $\mathcal{P}(q)$ are disjoint, and those in $\mathcal{P}(q)$ and $\mathcal{P}(q')$ with $q \neq q'$ lie in the disjoint sets U^q and $U^{q'}$. For (b), argue by induction on n , the case $n=0$ being trivial. A non-empty open $W \subseteq U_0$ meets D_n , hence $W \cap U^q \neq \emptyset$ for some $q \in \mathcal{P}_n$. The union of the

last entries of the positions in $\mathcal{P}(q)$ is dense in U^q , so that union meets $W \cap U^q$; since that union is contained in D_{n+1} , the set W meets D_{n+1} . Figure 3 shows the first two levels.

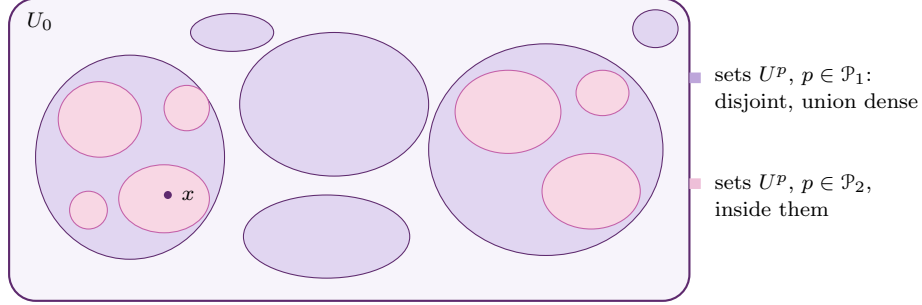


FIGURE 3. The proof of Theorem 2.3: the first two levels of the construction. Inside U_0 , the answers of σ at the first level form a maximal family of pairwise disjoint open sets, whose union D_1 is dense in U_0 ; inside each of them, the second level does the same. A point x lying in every D_n picks out exactly one set at each level, and these sets are the moves of Player I in a single play consistent with σ .

Each D_n is open and dense in U_0 , so $U_0 \setminus D_n$ is nowhere dense. Suppose that U_0 were of the second category. Then U_0 is not contained in the first-category set $\bigcup_n (U_0 \setminus D_n)$, so there is a point $x \in \bigcap_n D_n$. By (a), for each n there is exactly one $p_n \in \mathcal{P}_n$ with $x \in U^{p_n}$. By (c), the position p_{n+1} extends some $q \in \mathcal{P}_n$, and $x \in U^{p_{n+1}} \subseteq U^q$, so $q = p_n$ by (a). Thus the positions $p_0 \subseteq p_1 \subseteq \dots$ fit together into a single play consistent with σ , and $x \in \bigcap_n U^{p_n}$ lies in every set played by Player I. This contradicts the assumption that σ is winning. Hence U_0 is a non-empty open set of the first category, and the proof is complete. ☺



Remark 2.4. Part (2) does not say that Player II has a winning strategy in $\text{BM}(X)$ when X is a Baire space; in general neither player need have one (see [Tel87]).

3. THE STRONG CHOQUET GAME

In this section we add points to the moves of Player I, and show that the resulting game characterizes complete metrizable spaces (see Theorem 3.4 below). Here we prove the easy half of the theorem and derive from it the Baire category theorem (see Corollary 3.5 below); the hard half occupies Section 4.

Definition 3.1. Let X be a topological space. The *strong Choquet game* on X , denoted by $\text{Ch}(X)$, is played as follows. Player I chooses a pair (x_0, U_0) with $U_0 \in \mathcal{O}^+(X)$ and $x_0 \in U_0$. Player II responds with $V_0 \in \mathcal{O}^+(X)$ such that $x_0 \in V_0 \subseteq U_0$. At step $n \geq 1$, Player I chooses (x_n, U_n) such that $x_n \in U_n \subseteq V_{n-1}$, and Player II responds with $V_n \in \mathcal{O}^+(X)$ such that $x_n \in V_n \subseteq U_n$. Player II wins the play if $\bigcap_{n=0}^{\infty} U_n \neq \emptyset$.

In the format of Definition 1.2, Player I chooses from $X \times \mathcal{O}^+(X)$ and Player II from $\mathcal{O}^+(X)$. If the points x_n are omitted, what remains is exactly the Banach–Mazur game $\text{BM}(X)$, which in this context is also called the *Choquet game*. Following Choquet, who called the two players β (the first) and α (the second), the space X is called α -favorable if Player II has a winning strategy in $\text{BM}(X)$, and *strongly* α -favorable if Player II has a winning strategy in $\text{Ch}(X)$; Kechris [Kec95] calls such spaces *Choquet* and *strong Choquet* spaces.

Lemma 3.2. *If X is strongly α -favorable, then X is α -favorable.*

Proof. Let τ be a winning strategy for Player II in $\text{Ch}(X)$. In $\text{BM}(X)$, when Player I plays U_n , Player II picks any point $x_n \in U_n$ and answers with the set that τ gives in reply to (x_n, U_n) . The moves (x_n, U_n) are legal in $\text{Ch}(X)$, because $x_n \in U_n \subseteq V_{n-1}$, so the sets U_n played are those of a play of $\text{Ch}(X)$ consistent with τ , and their intersection is non-empty. ☺

Example 3.3. Player I wins $\text{Ch}(\mathbb{Q})$: he follows the strategy of Example 2.2(1), adding any point of each set he plays.

Theorem 3.4 (G. Choquet). *Let X be a metrizable space. Then X is completely metrizable if and only if X is strongly α -favorable.*

Proof. Suppose first that X is completely metrizable, and fix a compatible complete metric d on X . We define a strategy τ for Player II. When Player I plays (x_n, U_n) , Player II chooses r_n with $0 < r_n < 2^{-n}$ such that the closed ball $\overline{B}(x_n, r_n)$ is contained in U_n , and plays $V_n = B(x_n, r_n)$.

Consider a play consistent with τ . For $m > n$ we have $x_m \in U_m \subseteq V_n = B(x_n, r_n)$, so $d(x_m, x_n) < 2^{-n}$. Thus (x_n) is a Cauchy sequence, converging to some $x \in X$. For each n , the points x_m with $m > n$ lie in $\overline{B}(x_{n+1}, r_{n+1})$ (since $x_m \in V_{n+1}$ for $m > n+1$), so $x \in \overline{B}(x_{n+1}, r_{n+1}) \subseteq U_{n+1} \subseteq U_n$. Hence $x \in \bigcap_n U_n$, and τ is a winning strategy for Player II (see Figure 4).

The converse, that a strongly α -favorable metrizable space is completely metrizable, is proved in Section 4. ☺

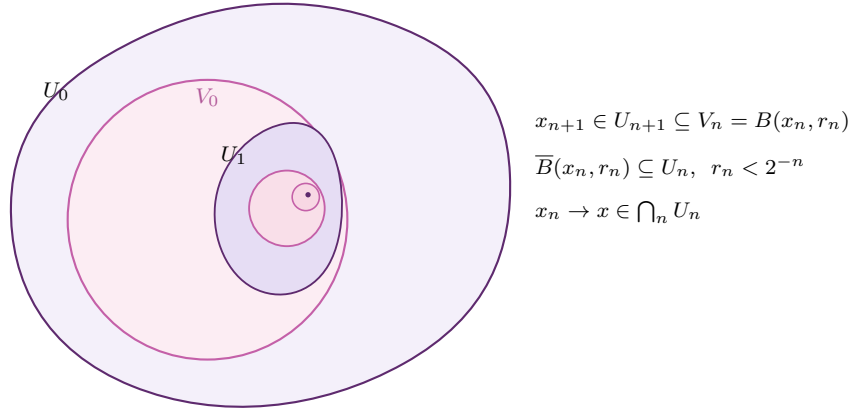


FIGURE 4. The easy half of Theorem 3.4. Player II answers each point x_n with a ball about it whose closure lies inside the set just played and whose radius is less than 2^{-n} . The points x_n form a Cauchy sequence, and its limit lies in every closed ball, hence in every U_n .

Corollary 3.5 (Baire category theorem). *Every completely metrizable space is a Baire space.*

Proof. Let X be completely metrizable. By Theorem 3.4 and Lemma 3.2, Player II has a winning strategy in $\text{BM}(X)$. Then Player I has none, and X is a Baire space by Theorem 2.3. ☺



Remark 3.6. The points x_n cannot be dropped: an α -favorable metrizable space need not be completely metrizable. Let $X = \{(a, b) \in \mathbb{R}^2 : b > 0\} \cup (\mathbb{Q} \times \{0\})$. The open upper half plane H is dense in X , and Player II wins $\text{BM}(X)$ by always answering with an open disk whose closure lies in $U_n \cap H$ and whose radius is less than 2^{-n} . But X is not completely metrizable: a closed subspace of a completely metrizable space is completely metrizable, and hence a Baire space by Corollary 3.5, whereas the closed subspace $\mathbb{Q} \times \{0\}$ of X is not a Baire space. By Theorem 3.4, Player II has no winning strategy in $\text{Ch}(X)$. In that game Player I can choose his points on the axis, and Player II must answer with sets containing them; Figure 5 shows how Player I then wins.

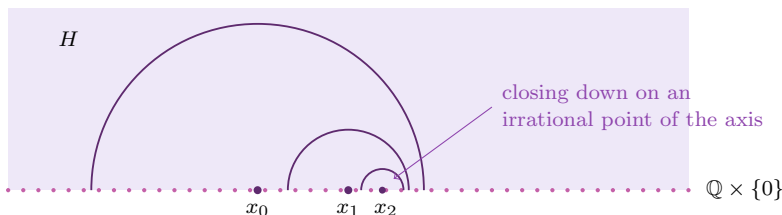


FIGURE 5. Player I wins $\text{Ch}(X)$ for the space X of Remark 3.6. He plays rational points of the axis, with small balls of X about them, inside the previous answer of Player II, and avoids one more rational each time. The balls close down on a point of the axis that is not rational, hence not in X .

4. THE CONVERSE OF CHOQUET'S THEOREM

In this section we prove the harder half of Theorem 3.4: if X is metrizable and Player II has a winning strategy in $\text{Ch}(X)$, then X is completely metrizable. The proof is long, so we first explain where a complete metric can come from and what the strategy has to do with it.

Where to look. Fix a compatible metric d on X . Every metric space is a dense subspace of a complete metric space, its *completion* $\tilde{X}$. If X is complete, then $\tilde{X} = X$; in general the completion contains new points, and the question is how X sits inside it. A countable intersection of open sets is called a G_δ set. Two subspaces of $\mathbb{R}$ show the difference. The irrationals form the set $\bigcap_{q \in \mathbb{Q}} (\mathbb{R} \setminus \{q\})$, a countable intersection of open sets, and they are completely metrizable (for instance, by continued fractions they are homeomorphic to the space of sequences of positive integers). The rationals are not a countable intersection of open subsets of $\mathbb{R}$: such an intersection would be a dense G_δ , and so would the irrationals, and the two disjoint dense G_δ sets would contradict the Baire category theorem. In Lemma 4.2 below we show that every G_δ subset of a complete metric space is completely metrizable. Our goal is therefore to find open sets $G_1, G_2, \dots$ of $\tilde{X}$ with $X = \bigcap_n G_n$.

Where the open sets come from. The strategy of Player II is a machine that answers points with small open sets. At the first stage, Player I may play any point x of X (with the open set X), and the strategy answers with an open set $V \ni x$. These sets cover X , and slightly enlarged into $\tilde{X}$, their union is an open set $G_1 \supseteq X$. At the second stage, Player I continues each of these positions with each point of the set just played, and so on. The open set G_n is, roughly, the union of the (enlarged) sets played by Player II at stage n , over all positions, so $X \subseteq G_n$ for every n . Suppose now that a point y of the completion lies in every G_n . If the sets containing y at the various stages all came from *one* play, then, since Player II

wins, that play would have a common point, and, since we shall arrange the sets to shrink, y would be that point; in particular, y would belong to X .

The difficulty. At each stage, y may lie in sets coming from many different positions, and choosing one position at each stage does not, in general, produce a single play: the position chosen at stage $n + 1$ must extend the one chosen at stage n . If at every stage only *finitely* many positions are involved, a compactness argument, König's lemma (see Lemma 4.4 below), makes the choices fit together. The finiteness is supplied by a theorem of A. H. Stone on metric spaces (see Lemma 4.3 below): every open cover can be refined to one in which each point has a neighborhood meeting only finitely many sets. Figure 6 shows how the two ingredients combine.

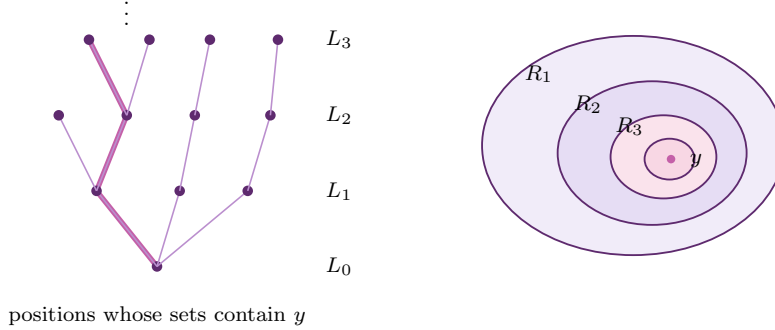


FIGURE 6. The end of the proof of Theorem 3.4. Left: for a point y of the completion lying in every G_n , the set L_n of positions of stage n whose sets contain y is finite and non-empty, and each member of L_{n+1} extends a member of L_n . König's lemma finds a branch (pink), that is, a single play. Right: along that play the sets R_n containing y shrink to a point; since Player II wins, the play has a common point in X , which must be y .

4.1. Four tools. Throughout this subsection, (X, d) is a metric space, and $\tilde{X}$ is a metric space, with metric $\tilde{d}$ extending d , in which X is dense. Closures and diameters of subsets of $\tilde{X}$ are taken in $\tilde{X}$. The first tool enlarges open sets of X to open sets of $\tilde{X}$ without making them bigger in diameter (see Figure 7).

Lemma 4.1. *For an open set $V \subseteq X$, put $\tilde{V} = \tilde{X} \setminus \overline{X \setminus V}$. Then the following hold:*

- (1) *the set $\tilde{V}$ is open in $\tilde{X}$, and $\tilde{V} \cap X = V$;*
- (2) *if $V \subseteq V'$, then $\tilde{V} \subseteq \tilde{V}'$;*
- (3) *the diameters satisfy $\text{diam } \tilde{V} = \text{diam } V$.*

Proof. (1) The set $\tilde{V}$ is the complement of a closed set. If $x \in V$, then, since V is open in X , some ball $B(x, \varepsilon)$ of X lies in V , so the ball of radius ε about x in $\tilde{X}$ misses $X \setminus V$, and $x \in \tilde{V}$. If $x \in X \setminus V$, then $x \in \overline{X \setminus V}$, and $x \notin \tilde{V}$.

(2) If $V \subseteq V'$, then $X \setminus V' \subseteq X \setminus V$, and the complements of the closures reverse the inclusion.

(3) Let $y \in \tilde{V}$ and $\varepsilon > 0$ be given. The set of points of $\tilde{V}$ at distance less than ε from y is open and non-empty, so it meets the dense set X , at a point of $\tilde{V} \cap X = V$. Thus $\tilde{V} \subseteq \overline{V}$, and $\text{diam } \tilde{V} \leq \text{diam } \overline{V} = \text{diam } V$. The reverse inequality holds because $V \subseteq \tilde{V}$. ☺

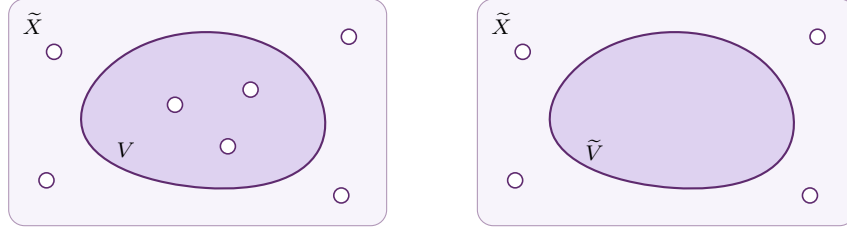


FIGURE 7. Lemma 4.1. The white dots are points of $\tilde{X}$ that are not in X . Left: an open set V of X , which misses the white dots inside it. Right: the open set $\tilde{V}$ of $\tilde{X}$, which also contains them. The two sets have the same points of X and the same diameter.

The second tool is the theorem of Alexandrov announced above; it reduces our problem to finding a G_δ set.

Lemma 4.2 (P. S. Alexandrov). *Every G_δ subset of a complete metric space is completely metrizable.*

Proof. Let (Y, ρ_0) be a complete metric space, and let $G = \bigcap_{n \geq 1} G_n$ with each G_n open in Y . We may assume that each closed set $F_n = Y \setminus G_n$ is non-empty, discarding the indices with $G_n = Y$ (if no index remains, then $G = Y$ and there is nothing to prove). The idea is to add to ρ_0 terms that blow up near the sets F_n , so that a Cauchy sequence cannot approach F_n . For $x \in G$ put $f_n(x) = 1/\rho_0(x, F_n)$, a continuous function on G , and for $x, y \in G$ put

$$\rho(x, y) = \rho_0(x, y) + \sum_{n \geq 1} 2^{-n} \min\{1, |f_n(x) - f_n(y)|\}.$$

Each term satisfies the triangle inequality, so ρ is a metric on G .

The metric ρ is compatible with the topology of G . Indeed, $\rho \geq \rho_0$, so ρ -convergence implies ρ_0 -convergence. Conversely, if $x_k \rightarrow x$ in G with respect to ρ_0 , then each term of the series tends to 0 by continuity of f_n , the series is dominated by $\sum_n 2^{-n}$, and hence $\rho(x_k, x) \rightarrow 0$.

The metric ρ is complete. Let (x_k) be ρ -Cauchy. Then (x_k) is ρ_0 -Cauchy, so it converges to some $y \in Y$. For each n , the sequence $(f_n(x_k))_k$ is Cauchy, hence bounded, say by c_n , so $\rho_0(x_k, F_n) \geq 1/c_n$ for every k , and therefore $\rho_0(y, F_n) \geq 1/c_n > 0$. Thus $y \in G_n$ for every n , that is, $y \in G$, and $\rho(x_k, y) \rightarrow 0$ by the compatibility just proved. ☺

The third tool supplies the finiteness. A family $\mathcal{V}$ of subsets of a space Y , indexed by a set I , is *locally finite* if every point of Y has a neighborhood that meets V_i for only finitely many indices $i \in I$. A family $\mathcal{V}$ *refines* a family $\mathcal{U}$ if every member of $\mathcal{V}$ is contained in some member of $\mathcal{U}$.

Lemma 4.3 (A. H. Stone). *Every open cover of a metric space has a locally finite open refinement that covers the space.*

The proof below is due to M. E. Rudin. The idea is to cover the space by small balls, assigning each ball to the first member of the cover that contains it generously, and to stop using small balls around points that are already covered by larger ones.

Proof. Let (Y, ρ) be a metric space, write $B(c, r)$ for open balls, and let $\{U_s : s \in S\}$ be an open cover of Y . By the well-ordering theorem we may assume that S is well ordered. For $s \in S$ and $n \geq 1$, let $V_{s,n}$ be the union of the balls $B(c, 2^{-n})$ over all centers $c \in Y$ such that:

- (i) the index s is the least one with $c \in U_s$;

- (ii) the center c lies in no set $V_{t,j}$ with $t \in S$ and $j < n$;
- (iii) the ball $B(c, 3 \cdot 2^{-n})$ is contained in U_s .

The definition is a recursion on n , because (ii) involves only the sets with smaller second index. Each $V_{s,n}$ is open, and $V_{s,n} \subseteq U_s$ by (iii).

The sets $V_{s,n}$ cover Y . Let $c \in Y$ be given, let s be the least index with $c \in U_s$, and choose n with $B(c, 3 \cdot 2^{-n}) \subseteq U_s$. Either $c \in V_{t,j}$ for some t and some $j < n$, or c satisfies (i)–(iii) and $c \in B(c, 2^{-n}) \subseteq V_{s,n}$.

The family is locally finite. Let $c \in Y$ be given, and choose s and n with $c \in V_{s,n}$. Choose $j \geq 1$ with $B(c, 2^{-j}) \subseteq V_{s,n}$. We show that the ball $B(c, 2^{-n-j})$ meets no set $V_{t,i}$ with $i \geq n+j$, and at most one set $V_{t,i}$ for each $i < n+j$; then it meets only finitely many sets of the family.

Let $i \geq n+j$, and let c' be a center used for $V_{t,i}$. Since $n < i$, condition (ii) gives $c' \notin V_{s,n}$, so $\rho(c, c') \geq 2^{-j}$. Every point of $B(c', 2^{-i})$ is therefore at distance at least $2^{-j} - 2^{-i} \geq 2^{-j} - 2^{-n-j} \geq 2^{-n-j}$ from c , and $B(c, 2^{-n-j})$ misses $V_{t,i}$.

Let $i < n+j$, and suppose that $p \in V_{t,i}$ and $q \in V_{t',i}$ with $t < t'$. Choose centers c_1 for $V_{t,i}$ and c_2 for $V_{t',i}$ with $\rho(p, c_1) < 2^{-i}$ and $\rho(q, c_2) < 2^{-i}$. By (iii), $B(c_1, 3 \cdot 2^{-i}) \subseteq U_t$, and by (i), $c_2 \notin U_t$ because $t < t'$. Hence $\rho(c_1, c_2) \geq 3 \cdot 2^{-i}$ and

$$\rho(p, q) \geq \rho(c_1, c_2) - \rho(p, c_1) - \rho(q, c_2) > 2^{-i} \geq 2 \cdot 2^{-n-j}.$$

Two points of $B(c, 2^{-n-j})$ are at distance less than $2 \cdot 2^{-n-j}$, so the ball cannot meet both $V_{t,i}$ and $V_{t',i}$. ☺

The last tool turns finitely many choices at each stage into one coherent choice.

Lemma 4.4 (D. König). *Let $L_0, L_1, \dots$ be non-empty finite sets, and for each n let $\pi_n: L_{n+1} \rightarrow L_n$ be a function. Then there are $\ell_n \in L_n$, for $n \geq 0$, with $\pi_n(\ell_{n+1}) = \ell_n$ for every n .*

Proof. For $m \geq n$ write $\pi_{n,m}: L_m \rightarrow L_n$ for the composite $\pi_n \circ \dots \circ \pi_{m-1}$ (the identity when $m = n$). Call $\ell \in L_n$ good if for every $m \geq n$ some element of L_m is mapped to ℓ by $\pi_{n,m}$. If $\ell \in L_n$ is the image of an element of L_m , then ℓ is also the image of an element of L_k for every k with $n \leq k \leq m$. Hence ℓ is good as soon as it is the image of elements of L_m for infinitely many m .

Some element of L_0 is good: every L_m is non-empty, so each m supplies an element of L_0 in the image of $\pi_{0,m}$, and since L_0 is finite, one element is supplied by infinitely many m . If $\ell \in L_n$ is good, then some element of the finite set $\pi_n^{-1}(\ell)$ is good: for each $m > n$, an element of L_m mapped to ℓ passes through some element of $\pi_n^{-1}(\ell)$, and one element of $\pi_n^{-1}(\ell)$ serves infinitely many m . Choosing recursively a good ℓ_0 and, for each n , a good $\ell_{n+1} \in \pi_n^{-1}(\ell_n)$ proves the lemma. ☺

4.2. The proof.

Proof of the converse of Theorem 3.4. Let X be metrizable, and let τ be a winning strategy for Player II in $\text{Ch}(X)$. Fix a compatible metric d on X and the completion $(\tilde{X}, \tilde{d})$ of (X, d) , in which X is dense, and use the notation $\tilde{V}$ of Lemma 4.1. We write $B(x, r)$ for the open ball of radius r about x in X . A *position of stage n* is a position of length $2n$ of $\text{Ch}(X)$ consistent with τ ; it ends with the move V_{n-1} of Player II when $n \geq 1$, and it is the empty position when $n = 0$.

Step 1: the families $\mathcal{S}_n$. By recursion on $n \geq 0$ we construct families $\mathcal{S}_n$ of pairs (R, p) , where R is a non-empty open subset of $\tilde{X}$ and p is a position of stage n . Write G_n for the union of the sets R with $(R, p) \in \mathcal{S}_n$. We require the following:

- (P1) if $n \geq 1$ and $(R, p) \in \mathcal{S}_n$, then $R \subseteq \tilde{V}$, where V is the last move of p , and $\text{diam } R \leq 2^{-n}$;
- (P2) the set G_n contains X ;
- (P3) the family of sets R , $(R, p) \in \mathcal{S}_n$, is locally finite in G_n ;
- (P4) each $(R, p) \in \mathcal{S}_{n+1}$ has a designated *parent* $(R', p') \in \mathcal{S}_n$ such that $R \subseteq R'$ and p extends p' by two moves: Player I plays $(x, R' \cap X)$ for some point x , and Player II answers as τ prescribes.

Put $\mathcal{S}_0 = \{(\tilde{X}, \emptyset)\}$; conditions (P2) and (P3) hold trivially.

Suppose that $\mathcal{S}_n$ has been constructed. Let $(R', p') \in \mathcal{S}_n$ and $x \in R' \cap X$ be given. The move $(x, R' \cap X)$ is legal for Player I at p' : for $n = 0$ it is a first move, and for $n \geq 1$ condition (P1) and Lemma 4.1(1) give $R' \cap X \subseteq \tilde{V} \cap X = V$, where V is the last move of p' . Let V_x be the answer of τ , so that $x \in V_x \subseteq R' \cap X$, let q_x be the resulting position of stage $n + 1$ (the sets V_x , the positions q_x and the sets O_x below depend also on the pair (R', p') , which we suppress from the notation), and put

$$O_x = R' \cap (V_x \cap B(x, 2^{-n-2}))^\sim,$$

where $W^\sim$ stands for $\tilde{W}$. The set O_x is open in $\tilde{X}$, contains x , is contained in R' and in $\tilde{V}_x$ (see Lemma 4.1), and has diameter at most 2^{-n-1} by Lemma 4.1(3). By (P2), every point of X lies in some R' , so the sets O_x , over all pairs $(R', p') \in \mathcal{S}_n$ and all $x \in R' \cap X$, cover X . Their union H is open in $\tilde{X}$, and the sets O_x form an open cover of the metric space H . By Lemma 4.3, this cover has a locally finite open refinement $\mathcal{R}$ that covers H ; we discard its empty members. For each $R \in \mathcal{R}$ choose a set O_x containing it, coming from the pair (R', p') and the point x , and put (R, q_x) into $\mathcal{S}_{n+1}$, with designated parent (R', p') .

Conditions (P1)–(P4) hold for $n + 1$. Indeed, $R \subseteq O_x \subseteq \tilde{V}_x$, where V_x is the last move of q_x , and $\text{diam } R \leq \text{diam } O_x \leq 2^{-n-1}$; the union G_{n+1} equals H , which contains X ; the family $\mathcal{R}$ is locally finite in $H = G_{n+1}$; and $R \subseteq O_x \subseteq R'$, with q_x extending p' by the moves $(x, R' \cap X)$ and V_x .

Step 2: a G_δ set containing X . The set $G = \bigcap_{n \geq 1} G_n$ is a G_δ subset of $\tilde{X}$, and $X \subseteq G$ by (P2). It remains to be shown that $G \subseteq X$.

Step 3: every point of G is the common point of a play. Let $y \in G$ be given. For $n \geq 0$, let L_n be the set of pairs $(R, p) \in \mathcal{S}_n$ with $y \in R$. Each L_n is non-empty, because $y \in G_n$ (and L_0 is the single pair $(\tilde{X}, \emptyset)$). Each L_n is finite, by (P3): some neighborhood of y meets only finitely many of the sets R , and every R containing y is among them. By (P4), the parent of a member of L_{n+1} contains y , so it lies in L_n ; let $\pi_n: L_{n+1} \rightarrow L_n$ be the parent map. By Lemma 4.4, there are $(R_n, p_n) \in L_n$ such that (R_n, p_n) is the parent of (R_{n+1}, p_{n+1}) for every n .

By (P4), each position p_{n+1} extends p_n by a move $(x_n, R_n \cap X)$ of Player I and the answer of τ . Hence the positions p_n fit together into a single play of $\text{Ch}(X)$, consistent with τ , in which Player I plays the sets $R_n \cap X$. Since τ is winning, there is a point $z \in \bigcap_n (R_n \cap X)$. For every $n \geq 1$, the points y and z both lie in R_n , which has diameter at most 2^{-n} by (P1). Hence $\tilde{d}(y, z) = 0$, and $y = z \in X$ (see Figure 6).

Step 4: conclusion. By Steps 2 and 3, we have $X = G$, a G_δ subset of the complete metric space $\tilde{X}$; since $\tilde{d}$ extends d , the topology that X inherits from $\tilde{X}$ is its own. By Lemma 4.2, the space X is completely metrizable. ☺



Remark 4.5. The points x_n of Player I are what make the proof work: at every stage, Player I plays every point of X , and this is why each G_n contains all of X . Without the

points the theorem fails (see Remark 3.6). For separable spaces, a different proof is given in [Kec95, Section 8].

APPENDIX A. HISTORICAL REMARKS

Baire category. Theorem 0.3 is an elementary reformulation of the definition and carries no attribution. This theorem should not be confused with the *Banach category theorem* [Ban30], which is deeper: in any topological space, the union of an arbitrary family of open sets of the first category is again of the first category. The title of Oxtoby's paper [Oxt57] refers to this theorem.

The Banach–Mazur and strong Choquet games. The Banach–Mazur game is Problem 43 of the Scottish Book, posed by S. Mazur in 1935 [Mau81]: two players alternately choose nested intervals of the real line, and the first player wins if their intersection meets a given set E . Mazur observed that the first player wins if E is comeager in some interval and that the second player wins if E is of the first category, and conjectured the converses. Banach recorded on 4 August 1935 that the conjecture is true, but his proof was never published. The first published proof is Oxtoby's [Oxt57]. He showed that in an arbitrary topological space the second player has a winning strategy exactly when E is of the first category; the other converse, that the first player wins exactly when E is comeager in some non-empty open set, requires further hypotheses on the space, and holds for instance in complete metric spaces. In the game $\text{BM}(X)$ of these notes the roles are arranged so that the target set is X itself: Player I plays the part of Mazur's second player on the open set U_0 he chooses first, and Theorem 2.3 is Oxtoby's theorem applied locally. See [Tel87] for a survey.

The strong Choquet game goes back to Choquet's work on Baire spaces [Cho58], and is expounded, together with Theorem 3.4, in his *Lectures on Analysis* [Cho69], where the players are called β and α . Kechris states Theorem 3.4 for separable metrizable spaces, where the theorem characterizes the Polish spaces [Kec95, Section 8]. The proof of Section 4, for arbitrary metrizable spaces, rests on three classical results. Lemma 4.2 is due to Alexandrov [Ale24]. Lemma 4.3 is A. H. Stone's theorem that metric spaces are paracompact [Sto48]; the short proof given here is M. E. Rudin's [Rud69]. Lemma 4.4 is Kőnig's infinity lemma [Kön27], first stated for graphs.

The notation $[\mathbb{N}]^\omega$ and the partition calculus. The companion notes on Ramsey theory play the Banach–Mazur game of Section 2 on the Ellentuck space of infinite subsets of $\mathbb{N}$, whose notation differs from that of Section 1; this paragraph explains the notation. For a set S and a cardinal κ , one writes $[S]^\kappa = \{X \subseteq S : |X| = \kappa\}$ and $[S]^{<\kappa} = \{X \subseteq S : |X| < \kappa\}$; thus $[\mathbb{N}]^\omega$ and $[\mathbb{N}]^{<\omega}$ are the families of infinite and of finite subsets of $\mathbb{N}$, as opposed to the sequence spaces A^ω and $A^{<\omega}$ of Section 1. This is the notation of the partition calculus of P. Erdős and R. Rado [ER56], in which $\kappa \rightarrow (\lambda)_\mu^r$ means: for $|S| = \kappa$, every partition of $[S]^r$ into μ pieces has a set $H \subseteq S$ with $|H| = \lambda$ and $[H]^r$ inside a single piece. Ramsey's theorem becomes $\omega \rightarrow (\omega)_k^r$ for finite r and k . For the infinite exponent the relation fails: using the axiom of choice, Erdős and Rado partitioned $[\mathbb{N}]^\omega$ into two pieces with no infinite homogeneous set [ER52]. Positive results recover $\omega \rightarrow (\omega)_2^\omega$ for definable partitions: Galvin and Prikry for Borel partitions [GP73], Silver for analytic ones [Sil70], and Ellentuck for partitions with the Baire property in the Ellentuck topology [Ell74]; in the companion notes on Ramsey theory, the Banach–Mazur game on the Ellentuck space is the bridge to these theorems. The partition calculus, and the school of infinite combinatorics that Erdős and

his collaborators built around this calculus, are set out in the monograph of Erdős, Hajnal, Máté, and Rado [EHMR84].

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